
Is There a Universal ‘Sounds Like AI’ Direction? Probing and Steering AI Style in Transformer Residual Streams

Joanna Zhang
joannazhang@uchicago.edu

Abstract

Detection of AI-generated text has become increasingly important as large language models are deployed at scale. Prior work has shown that high-level concepts such as truthfulness, refusal, and sentiment are encoded as linear directions in the residual streams of transformer models. We investigate whether “sounding like AI” — the composite stylistic quality that makes LLM outputs identifiable — is similarly represented, and whether steering along this direction can evade real-world AI detectors. Using contrastive activation addition (CAA) on Qwen2.5-3B, we extract an “AI style direction” from the HC3 dataset. Replicating prior work, we confirm that this direction achieves 97.5% classification accuracy on ChatGPT-sourced text, though 93% of the signal overlaps with text length. After controlling for length, an 85.5% accurate residual style direction remains. Steering along this direction reduces Pangram’s AI probability from 0.70 to 0.02 on ChatGPT-sourced text, demonstrating near-complete evasion of a state-of-the-art detector. We further extend the dataset to include Claude and Gemini responses, finding that these models produce substantially longer outputs (97.5% and 96.1% length overlap respectively), which largely dominates the extracted direction. To address this, we propose a sentence-boundary truncation method that reduces length-direction cosine similarity from 0.944 to 0.002, revealing a genuine cross-source style signal at 95.8% accuracy. However, steering on this length-controlled combined dataset produces only partial evasion (0.26 at most human-steered vs. 0.02 for ChatGPT-only), suggesting that while a cross-source AI style direction exists and is steerable, it is more diffuse than a single-source direction. Our results highlight both the promise and the limitations of residual stream manipulation as an approach to AI detection evasion.

1 Introduction

The proliferation of large language models has intensified an ongoing arms race between AI text generation and AI text detection. Tools such as GPTZero, ZeroGPT, and Pangram attempt to identify machine-generated text by exploiting statistical and stylistic signatures in LLM outputs, while a parallel line of work explores evasion strategies. Existing evasion approaches largely operate at the surface level: recursive paraphrasing Sadasivan et al. [2023], post-hoc humanization, or prompt engineering Mitchell et al. [2023]. These methods treat the model as a black box, modifying outputs after the fact without any understanding of *why* the text sounds AI-generated in the first place. We ask a more fundamental question: is “sounding like AI” encoded as an internal representation within the model itself, and if so, can we manipulate that representation directly to evade detection?

A growing body of work in mechanistic interpretability has established that transformer language models linearly encode high-level concepts as directions in their residual streams Marks and Tegmark [2024], Ardit et al. [2024], Konen et al. [2024], and that these directions can be extracted and causally

steered via Contrastive Activation Addition Rimsky et al. [2024]. Whether “sounding like AI” — the composite stylistic quality that makes LLM outputs identifiable — constitutes such a direction has not been directly investigated. Concurrently, Kuznetsov et al. (2025) use Sparse Autoencoders on residual streams to identify features contributing to AI text detection, finding that style and verbosity features are the most discriminative. Our work is complementary but distinct: we use the simpler difference-in-means approach and target *evasion* rather than detection interpretability.

We extract an “AI style direction” from Qwen2.5-3B using the HC3 dataset of paired human and ChatGPT responses, replicating prior work showing 97.5% classification accuracy with 93% overlap with text length. After controlling for length, a genuine residual style signal persists at 85.5% accuracy. Critically, steering along this direction reduces Pangram’s AI probability from 0.70 to 0.02, demonstrating near-complete evasion of a state-of-the-art detector through internal representation manipulation. We then extend to Claude and Gemini, finding that these models’ substantially greater verbosity dominates the extracted direction even more severely. To address this, we introduce a sentence-boundary truncation method that eliminates the length confound by construction, revealing a genuine cross-source style signal at 95.8% accuracy. Steering upon this dataset produces partial evasion (0.26 at most human-steered vs. 0.02 for ChatGPT-only), suggesting a cross-source AI style direction is steerable but more diffuse than a single-source direction.

Our main contributions are:

- We demonstrate near-complete evasion of Pangram (0.70 $\rightarrow$ 0.02 AI probability) via activation steering on ChatGPT-sourced text, establishing residual stream manipulation as a viable attack vector against AI detectors.
- We extend the analysis to three AI sources (ChatGPT, Claude, Gemini), finding that verbosity accounts for 93–97.5% of the extracted direction across all sources and is substantially more severe for Claude and Gemini, suggesting that current AI style directions are largely verbosity detectors.
- After controlling for length via sentence-boundary truncation, we find a genuine cross-source style signal at 95.8% accuracy, and show that this direction produces genuine but partial evasion (0.26 vs. 0.02 for ChatGPT-only) — suggesting AI style direction is source-specific in strength, with ChatGPT’s stylistic signal being more steerable than a pooled cross-source direction
- We find that CAA steering effectiveness decreases with model size across the Qwen2.5-Instruct family, with the 3B model reaching 0.219 Pangram probability vs. 0.741 and 0.605 for 7B and 14B respectively.

2 Background

We build directly on a preliminary study Hypogenic-AI [2024] that applied CAA to AI text detection, which we summarize here as it serves as our baseline.

Dataset. The study used HC3 Guo et al. [2023], a corpus of paired human and ChatGPT responses to the same questions across five domains. After filtering to 50–1500 characters, 200 pairs were used for direction extraction, 50 for validation, and 100 for test. ChatGPT responses are systematically longer than human responses (mean 914 vs. 446 characters), a confound that substantially affects the analysis.

Method. For each layer ℓ of Qwen 2.5-3B, a difference-in-means direction is computed between AI and human last-token residual stream activations. This direction serves both as a linear classifier and as a CAA steering vector during generation.

Findings. The direction at Layer 21 achieves 97.5% test accuracy (AUC = 0.999), but has 0.93 cosine similarity with a text length direction — it is almost entirely a verbosity detector. After projecting out length, a residual style direction still achieves 85.5% accuracy at Layer 33. Causal steering qualitatively shifts output style but with modest effect: an LLM judge found only a ~ 1 point shift on a 7-point scale, attributed to the base model’s high AI-likeness floor.

87 3 Related Work

88 **Linear representations in residual streams.** A growing body of work in mechanistic interpretability
89 has shown that transformer language models encode high-level concepts as linear directions in their
90 residual streams. Marks and Tegmark [2024] provide foundational evidence, demonstrating that
91 truth and falsehood are linearly represented and extractable via difference-in-means probing — the
92 core technique we employ — with causal verification via surgical intervention in the forward pass.
93 Arditì et al. [2024] extend this to behavioral properties, showing that refusal in safety-tuned models
94 is mediated by a single direction whose ablation disables refusal and whose addition induces it
95 on harmless inputs. Rinsky et al. [2024] introduce Contrastive Activation Addition (CAA), the
96 methodology directly underlying our work, which extracts steering vectors by averaging residual
97 stream differences between contrastive pairs and applies them at inference time with a tunable
98 coefficient. Konen et al. [2024] further demonstrate that style properties — sentiment, emotion,
99 writing style — can be extracted and steered via activation engineering without any training. Our
100 work asks whether “sounding like AI” is similarly encoded, extending this line of research to a
101 stylistic property with direct practical implications for detection evasion.

102 **AI text detection and evasion.** Mitchell et al. [2023] introduce DetectGPT, a zero-shot detector
103 exploiting the observation that LLM-generated text occupies negative curvature regions of a model’s
104 log probability function. Sadasivan et al. [2023] stress-test the robustness of a broad range of detectors
105 — including watermarking, neural network-based, and zero-shot approaches — showing empirically
106 and theoretically that existing methods are unreliable under paraphrasing attacks. Our approach
107 is fundamentally different from such surface-level evasion: rather than modifying outputs after
108 generation, we intervene directly on the model’s internal style representation during the forward pass.

109 **Residual stream features for AI text detection.** Most closely related to our work, Kuznetsov et al.
110 [2025] apply Sparse Autoencoders (SAEs) to Gemma-2-2b’s residual stream to identify features
111 contributing to AI text detection, finding that style and verbosity features are the most discriminative.
112 They also find, consistent with our results, that text length is a major confounding factor. Our work
113 differs in three key ways: we use the simpler difference-in-means approach rather than SAEs; we
114 target evasion rather than detection interpretability; and we evaluate against real-world detectors
115 including Pangram.

116 4 Methodology

117 Our methodology extends the preliminary study’s pipeline along three axes: replacing the LLM judge
118 with real AI detectors, testing the instruct variant of the base model, and building a multi-source
119 dataset. We describe each in turn.

120 4.1 Models

121 We run all experiments on **Qwen2.5-3B** (base) and **Qwen2.5-3B-Instruct**, loaded in float16 precision.
122 For the model size experiment, we additionally run the instruct pipeline on Qwen2.5-7B-Instruct and
123 Qwen2.5-14B-Instruct. Activations are extracted from the last-token position of the residual stream
124 at all 37 layer outputs (embedding layer + 36 transformer layers), with a maximum input length of
125 512 tokens.

126 4.2 Direction Extraction and Classification

127 We follow the CAA methodology of the preliminary study exactly. For each layer ℓ , the AI direction
128 is computed as the difference-in-means over the training split:

$$\mathbf{d}_\ell = \frac{1}{N} \sum_{i=1}^N \mathbf{a}_\ell^{(\text{AI}, i)} - \frac{1}{N} \sum_{i=1}^N \mathbf{a}_\ell^{(\text{human}, i)}$$

129 Classification is performed by projecting held-out activations onto the normalized direction and
130 thresholding at the class midpoint. The best layer is selected by validation accuracy. We also compute
131 length-orthogonal directions by projecting out a separately computed length direction (derived from
132 a median split on character count) and sweeping across all layers for the best length-controlled
133 accuracy.

4.3 Multi-Source Dataset

To test whether the AI direction generalizes beyond ChatGPT, we collected responses from **Claude** (claude-sonnet-4-5) and **Gemini** (gemini-2.5-flash) to the same HC3 questions via OpenRouter, yielding approximately 350 pairs per source. We removed the upper character length filter from the preliminary study, as Claude and Gemini responses are substantially more verbose than ChatGPT (mean lengths of 1,245 and 1,741 characters respectively, vs. 939 for ChatGPT). We run the direction-finding pipeline separately for each source, and also construct a pooled dataset combining all three sources (704 pairs after filtering: 329 ChatGPT, 288 Claude, 87 Gemini) to find a shared AI direction with the human side held constant.

Source	Pairs	Human mean length (chars)	AI mean length (chars)
ChatGPT	350	506	939
Claude	350	506	1,245
Gemini	347	508	1,741
Combined	704	525	1,052
Combined (truncated)	1,031	513	501

Table 1: Dataset statistics per source. Human mean lengths are nearly identical across sources since all draw from the same HC3 questions; the large variation in AI mean length reflects differences in verbosity across models. The combined dataset (704 pairs) applies stricter filtering than the individual pipelines, disproportionately reducing Gemini pairs (329 ChatGPT, 288 Claude, 87 Gemini). The truncated combined dataset retains all 1,031 pairs (345/344/342 per source) by sentence-level truncation of AI responses to match human length, nearly eliminating the length gap.

4.4 AI Detector Evaluation

Rather than using an LLM judge, we evaluate steered outputs against three commercial detectors: **Pangram** (v3 API, ai_assistance_score), **GPTZero** (completely_generated_prob), and **ZeroGPT** (fakePercentage/100). All three return a scalar probability in $[0, 1]$. GPTZero was evaluated manually via web interface due to API cost. We evaluate five steering multipliers (two negative, baseline, two positive) across thirty prompts sampled from the HC3 test set for the base model and truncated combined dataset experiments, and across five fixed prompts for remaining experiments, scoring each generated output independently.

4.5 Prompting Baseline

To assess whether instruction prompting achieves comparable evasion to activation steering, we prepend one of three “write like a human” instructions to each prompt and score the outputs with Pangram. We run this on both the base and instruct models: the base model serves as a sanity check (Qwen2.5-3B does not reliably follow instructions), while the instruct model provides a fairer comparison against CAA steering.

4.6 Causal Steering

Steering is implemented via a forward hook registered on the target transformer layer. At each forward pass during generation, the hook adds $\alpha \cdot \hat{\mathbf{d}}_\ell$ to the residual stream at every token position, where α is the steering multiplier and $\hat{\mathbf{d}}_\ell$ is the normalized direction scaled by its raw magnitude. Generation uses temperature 0.7, top-p 0.9, and a maximum of 150 new tokens.

5 Experiments and Results

5.1 Direction Finding: Base Model Replication

We first replicate the preliminary study’s direction-finding pipeline on the HC3 ChatGPT dataset using Qwen2.5-3B. Figure 1 shows classification performance and PCA cluster separation across all 37 layers. The direction-finding results closely match the preliminary study: the best layer by

validation accuracy is Layer 21, with test accuracy of 97.5% (AUC = 0.999, 95% CI: [0.950, 0.995]). Cross-layer direction consistency is high (mean adjacent cosine similarity: 0.874), confirming that the direction stabilizes early and remains stable throughout the network. One notable feature is a consistent dip in classification accuracy around layers 8–10 before recovering sharply at layer 11, visible in both the accuracy and silhouette plots; the cause of this is unclear.

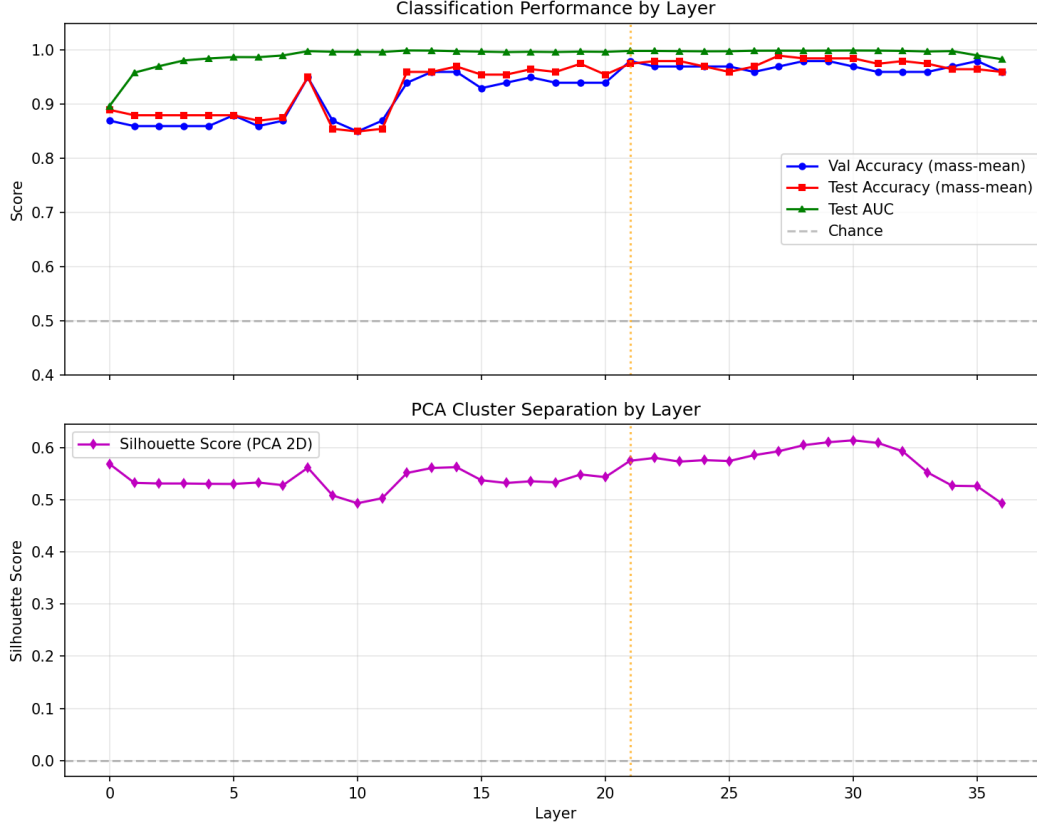


Figure 1: Classification performance (top) and PCA cluster separation (bottom) by layer for Qwen2.5-3B on the HC3 ChatGPT dataset. The orange dotted line marks the best layer (Layer 21). A consistent dip around layers 8–10 is visible in both metrics.

The length confound is substantial: cosine similarity between the AI direction and the length direction is 0.930, and accuracy drops from 97.5% to 82% after projecting out length. The best length-orthogonal direction is found at Layer 33 with 85.5% accuracy, consistent with the preliminary study.

5.2 Detector Evasion: Base Model

Replacing the LLM judge with Pangram yields a substantially stronger evasion result than the preliminary study suggested. Figure 2 shows Pangram AI probability across five steering multipliers. At the most human-steered condition (multiplier −33), Pangram assigns only 2% AI probability — effectively fooled. The baseline (multiplier 0) already scores 0.70, confirming that Qwen2.5-3B outputs are distinctly AI-sounding even without steering.

5.3 Instruct Model

Running the same pipeline on Qwen2.5-3B-Instruct yields a different best layer (Layer 24, test accuracy 97.5%, AUC = 0.998) and a different best length-orthogonal layer (Layer 19, 86% accuracy). Despite the instruct model having a more pronounced AI style — baseline Pangram score of 0.99 vs. 0.70 for the base model — steering is substantially less effective. The most human-steered condition

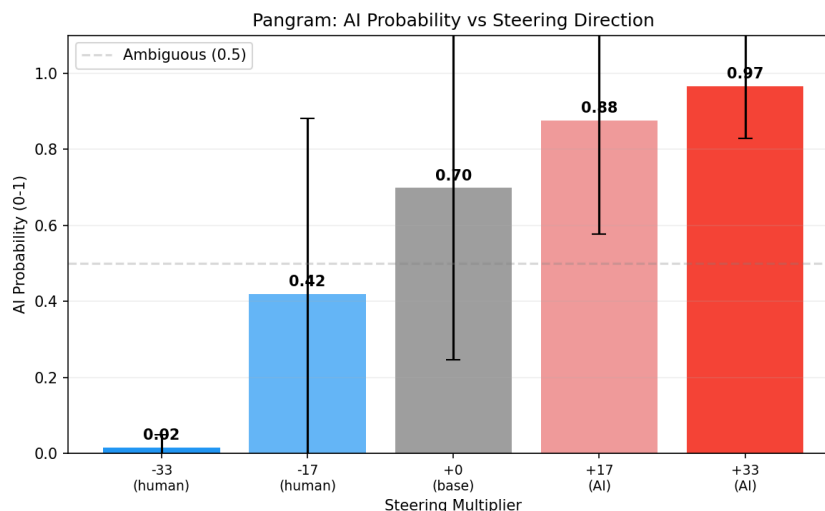


Figure 2: Pangram AI probability by steering multiplier for Qwen2.5-3B steered with the ChatGPT-derived AI direction. Negative multipliers steer toward human-like text; positive multipliers steer toward AI-like text. Error bars show standard deviation across 30 prompts.

187 only reaches 0.44, compared to 0.02 for the base model. The more entrenched AI style of the instruct
 188 model appears to make it harder to steer away from, rather than providing a larger dynamic range as
 189 initially hypothesized.

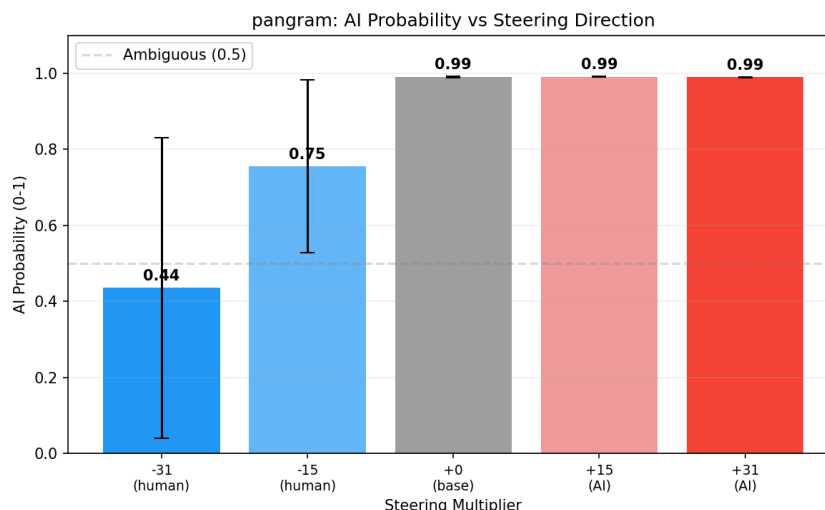


Figure 3: Pangram AI probability by steering multiplier for Qwen2.5-3B-Instruct. Compared to the base model (Figure 2), the baseline is higher (0.99 vs. 0.70) and the most human-steered condition reaches only 0.44 vs. 0.02, suggesting the instruct model’s stronger AI style is harder to override via activation steering.

190 5.4 Does the AI Direction Generalize Across Sources?

191 Table 2 shows direction-finding results across all three sources. All three achieve high classification
 192 accuracy, but the length confound is even more severe for Claude and Gemini than for ChatGPT.

193 After projecting out length, Claude and Gemini drop to 62–64% accuracy, barely above chance,
 194 compared to ChatGPT’s 80.5%. The within-class correlations tell the same story: Gemini’s correlation
 195 between AI text length and projection onto the AI direction is essentially 0 (0.004), meaning Gemini

Source	Best Layer	Test Acc	Test AUC	Cos sim w/ length	Acc (orig)	Acc (no length)
ChatGPT	22	98.5%	1.000	0.938	98.5%	80.5%
Claude	28	98.0%	0.999	0.975	98.0%	62.0%
Gemini	35	97.4%	0.996	0.961	97.4%	64.4%

Table 2: Direction-finding results per source. The length confound is severe across all sources, but especially for Claude and Gemini, whose length-orthogonal directions barely exceed chance.

196 responses are uniformly long regardless of the question. Claude is similar (0.060). This suggests that
197 for Claude and Gemini, the learned direction is almost entirely a verbosity detector with little residual
198 style signal.

199 Table 2 reports classification accuracy per source; Table 3 reports pairwise cosine similarities between
200 the extracted directions, both raw and length-orthogonal.

Pair	Cosine Sim (raw)	Cosine Sim (length-orthogonal)
ChatGPT vs. Claude	0.504	0.467
ChatGPT vs. Gemini	0.151	0.197
Claude vs. Gemini	0.430	0.251

Table 3: Pairwise cosine similarities between per-source AI directions, both raw (best layer per source) and after projecting out the length component. ChatGPT and Claude share moderate overlap; ChatGPT and Gemini are nearly orthogonal, suggesting their directions capture largely source-specific rather than universal style signals.

201 The low cross-source similarities (Table 3) are consistent with the evasion results in Figure 4:
202 ChatGPT and Claude share moderate directional overlap (0.50 raw), while ChatGPT and Gemini
203 are nearly orthogonal (0.15). Notably, the length-orthogonal similarities are lower for both Claude
204 pairs than the raw similarities, suggesting that once the shared verbosity signal is removed, what
205 remains is even more source-specific. This is reflected in the evasion results: ChatGPT’s direction
206 gives the cleanest evasion (dropping to 0.19 at most human), while Claude and Gemini directions
207 are far less effective (0.75 and 0.85 respectively). The baseline scores for Claude and Gemini are
208 both 0.99, reflecting their high verbosity. One anomaly is that Gemini’s “Most AI” condition actually
209 drops to 0.84, lower than its baseline — likely an artifact of incoherent outputs at high multipliers
210 being misclassified as human by Pangram.

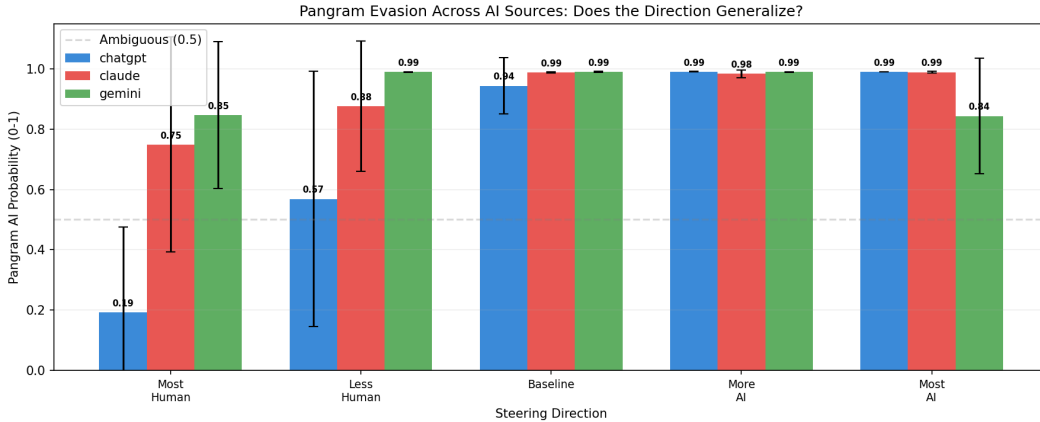


Figure 4: Pangram AI probability by steering direction and source. Each source is steered with its own best-layer direction. ChatGPT’s direction produces the cleanest evasion; Claude and Gemini directions show little effect, consistent with their directions being dominated by length.

211 5.5 Pooled Dataset

212 Combining all three sources into a pooled dataset (704 pairs) and finding a single shared direction
 213 gives a best layer of 33 with 97.1% test accuracy (AUC = 0.995). The length confound remains
 214 strong (cosine similarity 0.944), and accuracy drops from 97.1% to 70.1% after removing length —
 215 worse than ChatGPT alone but better than Claude or Gemini individually. Steering on the pooled
 216 direction produces a reversed result: negative multipliers (meant to steer human) score *higher* on
 217 Pangram than positive multipliers, the opposite of what the ChatGPT-only pipeline found. Inspecting
 218 individual outputs reveals that the effect is driven by a few outlier prompts dropping dramatically
 219 (e.g. to 0.005–0.067) while most prompts remain near 0.99 regardless of multiplier. We attribute
 220 the direction flip to the pooled “AI” centroid being pulled differently when mixing three sources
 221 with very different length distributions, potentially causing the mass-mean direction to point in the
 222 opposite orientation compared to the ChatGPT-only direction.

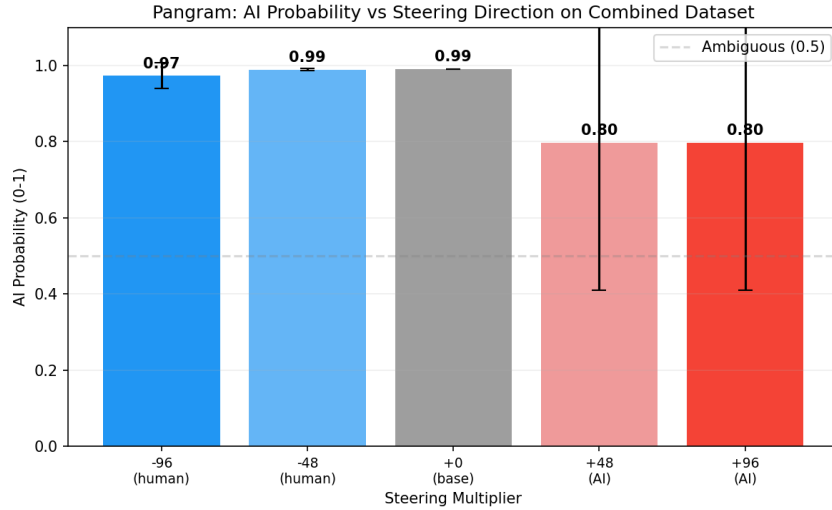


Figure 5: Pangram AI probability by steering multiplier on the pooled combined dataset (704 pairs, 5 prompts per condition). The direction is reversed relative to the ChatGPT-only result: human-steered conditions score higher (0.97–0.99) than AI-steered conditions (0.80), with high variance on the positive multipliers suggesting instability. As shown in Section 5.6, length-controlling the dataset via truncation partially resolves this reversal.

223 5.6 Truncation Experiment

224 To directly test whether a genuine style signal exists independent of length, we truncate AI responses
 225 sentence-by-sentence to match human response lengths (Section 4). This reduces the cosine similarity
 226 between the AI direction and the length direction from 0.944 to 0.002, and accuracy before and after
 227 removing length is identical (95.8%), confirming that the learned direction is now capturing style
 228 rather than verbosity. Direction finding on the truncated dataset gives a best layer of 29 with 95.8%
 229 test accuracy (AUC = 0.996, 95% CI: [0.935, 0.977]).

230 Figure 6 shows Pangram evasion results on the truncated dataset evaluated over 30 prompts. The
 231 direction flip observed in earlier 5-prompt evaluations is no longer present: steering now follows the
 232 expected monotonic pattern, with the most human-steered condition reaching 0.26 and the baseline
 233 at 0.78. This suggests the reversal in the 5-prompt results was a sampling artifact rather than a true
 234 property of the pooled direction. Evasion is less complete than the ChatGPT-only result (0.26 vs.
 235 0.02), consistent with the pooled direction capturing a noisier, more averaged style signal across three
 236 sources. Variance across prompts remains high, reflecting the heterogeneity of the combined dataset.

237 5.7 Prompting Baseline

238 Figure 7 shows the comparison between CAA steering and prompting baseline on the instruct model.
 239 All three instruction variants flatline at approximately 0.99 Pangram score, identical to the unsteered

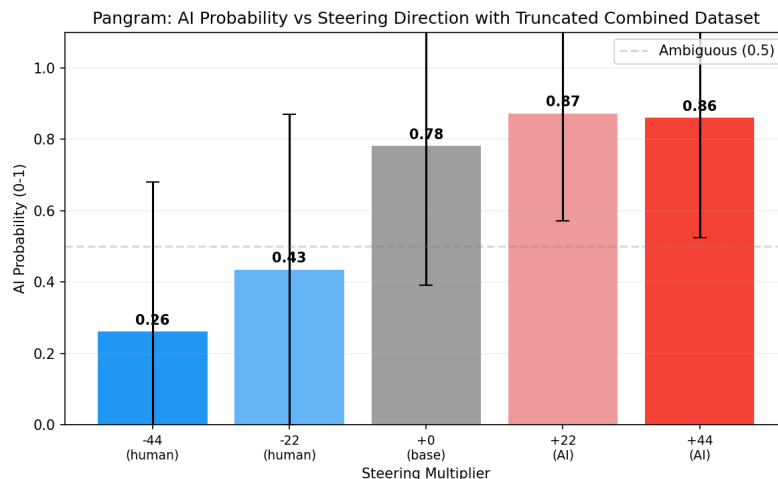


Figure 6: Pangram AI probability by steering multiplier on the truncated pooled dataset, evaluated over 30 prompts. The direction flip seen in earlier 5-prompt evaluations (see appendix) is absent; steering follows the expected monotonic pattern ($0.26 \rightarrow 0.43 \rightarrow 0.78 \rightarrow 0.87 \rightarrow 0.86$). Evasion is less complete than the ChatGPT-only result but genuine, with high variance reflecting the heterogeneity of the combined dataset.

240 baseline. CAA steering reduces the score to 0.44 at the most human-steered condition, substantially
 241 outperforming prompting. On the base model, prompting similarly has no effect, as expected given
 242 that Qwen2.5-3B does not follow instructions reliably.

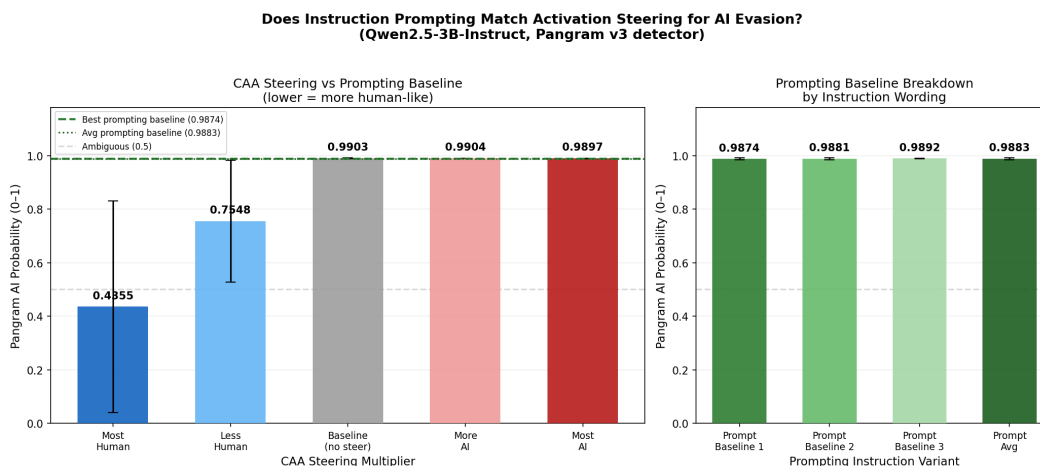


Figure 7: CAA steering vs. prompting baseline on Qwen2.5-3B-Instruct (Pangram v3). Left: steering multipliers with prompting baseline overlaid as dashed lines. Right: breakdown by instruction wording. All prompting variants flatline near 0.99; CAA steering achieves 0.44 at the most human-steered condition.

243 5.8 Effect of Model Size

244 To test whether model size affects the steerability of the instruct model, we ran the CAA pipeline on
 245 Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, and Qwen2.5-14B-Instruct using the ChatGPT-derived
 246 direction. Figure 7 shows Pangram AI probability across steering conditions for all three model sizes.
 247 All three models have identical baselines (0.993), but steering effectiveness decreases with model
 248 size: the 3B model reaches 0.219 at the most human-steered condition, while the 7B and 14B models
 249 only reach 0.741 and 0.605 respectively. The 14B model being slightly more steerable than 7B is

unexpected and may be an artifact of the small sample size (5 prompts per condition). The general trend suggests that larger instruct models are harder to steer away from AI-sounding outputs via activation addition, consistent with the hypothesis that larger models have more deeply entrenched stylistic patterns from RLHF fine-tuning.

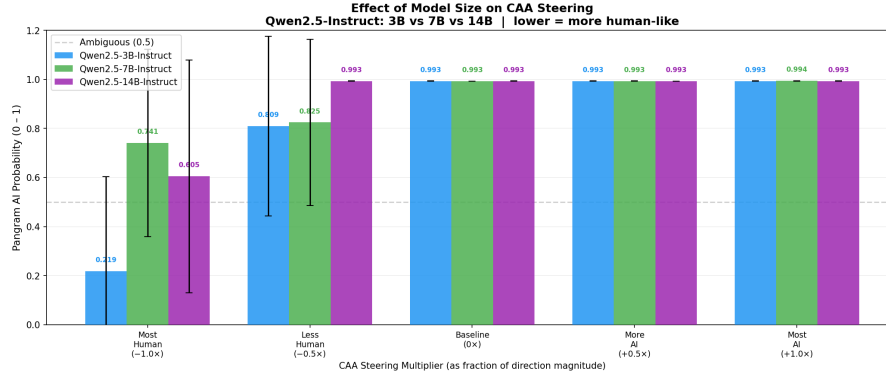


Figure 8: Effect of model size on CAA steering effectiveness (Pangram v3, 5 prompts per condition). All three Qwen2.5-Instruct models share an identical baseline of 0.993, but the 3B model is substantially more steerable than the 7B and 14B models. Note that results are based on only 5 prompts per condition and should be interpreted with caution.

6 Discussion & Conclusion

The length confound dominates, but a real style signal exists. The most consistent finding across all experiments is that the AI direction is overwhelmingly a verbosity detector. This is most extreme for Claude and Gemini, where over 96% of the learned direction aligns with text length, and length-orthogonal accuracy barely exceeds chance. The truncation experiment provides the clearest evidence that a genuine style signal exists beyond length: after eliminating the length gap, classification accuracy remains at 95.8%, indicating the model encodes real stylistic differences between AI and human text. However, this style signal is harder to exploit for evasion than the length-confounded direction.

Truncation partially recovers evasion. The per-source results show that ChatGPT’s direction produces the cleanest evasion (0.02), while Claude and Gemini’s own directions are largely ineffective (0.75 and 0.85 respectively), consistent with their directions being dominated by length. The pooled direction (704 pairs, no truncation) produces a reversed result: human-steered conditions score *higher* on Pangram (0.97–0.99) than AI-steered conditions (0.80), the opposite of the ChatGPT-only result. We attribute this to the pooled “AI” centroid being pulled differently when mixing three sources with very different length distributions, causing the mass-mean direction to point in the opposite orientation. Truncating AI responses to match human length (1031 pairs) partially resolves this: the direction now follows the expected monotonic pattern, achieving 0.26 probability at the most human-steered condition. This is better than either Claude or Gemini individually (without controlling for length) but weaker than ChatGPT alone, suggesting that truncating the pooled dataset recovers a shared style signal obscured by verbosity, but that ChatGPT’s stylistic signal remains the dominant contributor to cross-source evasion.

Unsurprisingly, activation steering outperforms prompting. The prompting baseline averages around 0.99 on both the base and instruct models with very little variation, while CAA steering achieves 0.02 on the base model and 0.44 on the instruct model. The base model result is somewhat unsurprising since Qwen2.5-3B does not follow instructions reliably, but the instruct model result is more telling: even with a model that can follow instructions, explicit prompting to “write like a human” has no measurable effect on Pangram scores. This suggests that whatever Pangram is detecting is not easily overridden by surface-level stylistic intent, but can be partially overridden by intervening directly on internal representations.

Limitations. Several limitations affect the strength of our conclusions. First, the majority of steering evaluations use only 5 prompts per condition, which is too few to draw robust conclusions — a single

incoherent or anomalous output can substantially shift the mean, as seen in the pooled and truncated dataset results. While initial evaluations used 5 prompts per condition due to API cost constraints, key experiments were subsequently rerun with 30 prompts; remaining experiments retain the 5-prompt evaluation. Second, GPTZero and ZeroGPT results were noisy and non-monotonic, possibly due to the short length of steered outputs (both detectors noted unreliability on inputs less than 250 words); we were unable to draw reliable conclusions from either detector. Finally, all experiments use a single model family (Qwen2.5); generalization to other architectures remains untested.

Conclusion. We extended a preliminary study on mechanistic interpretability of AI-generated text in three directions: multi-source dataset construction, real detector evaluation, and length control. Our results confirm that a linear AI direction exists in Qwen2.5-3B’s residual stream, and that steering with the ChatGPT-derived direction can substantially reduce Pangram detection scores. However, this direction does not generalize cleanly across AI sources, and a pooled multi-source direction fails to produce consistent evasion. The dominant confound throughout is text length, particularly for Claude and Gemini. Taken together, our findings suggest that “sounding like AI” is not a single universal direction in representation space, but a source-specific composite of style and verbosity that varies across model families.

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336 A Example Steered Outputs

337 Below are some examples of steering results, run on Qwen2.5-3B. For each prompt, we show five
338 steering conditions ranging from most human-like to most AI-like with their multipliers. Outputs are
339 truncated at the generation limit.

340 Prompt 1: Write a short paragraph about climate change:

341 **Most Human** (−33.2) The average temperature on Earth is rising. The earth’s atmosphere is getting
342 warmer. This is caused by the greenhouse effect. This greenhouse effect is the result of CO2
343 and other green house gases. This is causing the temperature to rise and rise. If we do not
344 reduce our green house gas emissions, the temperature will rise to 3 degrees C. The sea level
345 will rise 120 m. We will not be able to live on the planet. If we don’t reduce our green house
346 gas emissions, there will be a flood of 200 cm in London. The ice in the north and south
347 pole will melt and there will be a rise in sea level. The sea will flood our coasts. Our coasts
348 will [...]

349 **Less Human** (−16.6) Climate change is the long-term alteration of the statistical distribution of
350 weather patterns over several decades or longer. Climate change is a change in the statistical
351 distribution of weather over periods of time that exceed a few decades which is typically
352 associated with a change in the mean or the variability of those processes. It may be in
353 response to natural or human-induced changes in radiative forcing of the climate system, or
354 to changes in physical or biological systems. The changes in climate may be identified in
355 records of actual surface weather phenomena, in measurements of physical properties of the
356 atmosphere, or in the amount and distribution of snow and ice, or in sea level.

357 **Baseline** (0) Climate change is a global phenomenon that refers to the long-term alteration of Earth’s
358 climate patterns and weather conditions. It is primarily caused by human activities, such as
359 the burning of fossil fuels, deforestation, and industrial processes, which release greenhouse
360 gases into the atmosphere. These gases trap heat from the sun and contribute to the warming
361 of the planet. The consequences of climate change are far-reaching, including rising sea
362 levels, more frequent and intense natural disasters, and significant impacts on ecosystems
363 and biodiversity. Urgent action is needed to mitigate its effects and adapt to the changes
364 already occurring.

365 **More AI** (+16.6) Climate change refers to the long-term changes in weather patterns and temper-
366 ature that are caused by various factors, including human activities such as burning fossil
367 fuels and deforestation. These changes can have significant impacts on ecosystems, wildlife,
368 and human societies, such as rising sea levels, more frequent extreme weather events, and
369 disruptions to agriculture and water resources. Efforts to mitigate climate change involve
370 reducing greenhouse gas emissions, promoting renewable energy sources, and implementing
371 sustainable practices in various sectors of society.

372 **Most AI** (+33.2) Climate change is a pressing issue that has become increasingly important in
373 recent years. As the world continues to face the impacts of climate change, it is important
374 for individuals, governments, and organizations to take steps to reduce greenhouse gas
375 emissions and promote sustainability. This includes measures such as reducing reliance
376 on fossil fuels, investing in renewable energy sources, and promoting energy efficiency.
377 Additionally, individuals can make lifestyle changes such as reducing their carbon footprint
378 by conserving energy, using public transportation, and supporting environmentally-friendly
379 products.

380 Prompt 2: Tell me about the history of pizza:

381 **Most Human** (−33.2) Pizza is one of the most popular dishes in the world. Pizza is a round flatbread
382 baked in a pizza oven and topped with a variety of things. The flatbread is called a pizza
383 base. The pizza base is flat and round, and has a very thin base. The base is made of flour,
384 salt, yeast and water. The yeast is what makes the dough rise. It is like a balloon. The dough
385 is stretched into a circle. The dough is rolled up into a circle and baked. The baking makes
386 the dough very thin. The pizza base is then topped with cheese, ham, pepperoni, mushrooms,

387 olives, onions, whatever you like. The pizza is then cooked for a few minutes. When it is
388 cooked it [...]

389 **Less Human** (−16.6) Pizza originated in Naples, Italy, and was brought to the United States by
390 Italian immigrants. The first pizza restaurant in the US was opened in 1905 in New York.
391 The first pizza restaurant in London opened in 1919. In 1929, the first pizza parlor opened in
392 Canada. In 1936, pizza became an official meal in the US Army. In 1958, Domino's Pizza
393 was founded by Tom and Mary Simonsen. In 1960, the first pizza chain in the US opened.
394 In 1962, Pizza Hut was founded. In 1964, Papa John's opened. In 1975 [...]

395 **Baseline** (0) Pizza is a dish of Italian origin consisting of a usually round, flattened base of leavened
396 wheat-based dough topped with tomatoes, cheese, and various other ingredients, which is
397 usually baked in an oven. Pizza originated in Naples, Italy.

398 **More AI** (+16.6) Pizza is a popular dish that originated in Italy. It is believed to have been created
399 in Naples, Italy, during the 18th century. The first pizzas were made with simple ingredients
400 like tomato sauce, mozzarella cheese, and olive oil. Over time, different regions in Italy
401 developed their own unique styles of pizza, such as the Neapolitan-style pizza and the
402 Sicilian-style pizza. Pizza became popular throughout the world, and today it is enjoyed by
403 people of all ages and backgrounds.

404 **Most AI** (+33.2) Pizza is a popular and beloved dish that has a rich history dating back to ancient
405 times. The origins of pizza can be traced back to various regions around the world, but the
406 modern version of pizza as we know it today is typically associated with Italy. Here's a
407 brief overview of the history of pizza: 1. Ancient Origins: The roots of pizza can be traced
408 back to various cultures around the world, with some evidence suggesting that early forms
409 of flatbread, such as unleavened bread or flatbreads made from grains like wheat or barley,
410 were used in ancient times. 2. Early Italian Pizza: The term "pizza" comes from the Italian
411 word "pizza," which means "flatbread." The earliest versions of pizza in Italy were often [...]

412 **B Prompts**

413 Below, find the three prompts used in the prompting baseline:

- 414 1. "Write in a natural, conversational human style. Avoid sounding like an AI."
- 415 2. "Respond as a real person would: casually, directly, and without over-explaining."
- 416 3. "Write like a human, not like an AI assistant. Be informal and natural."

417 **C Additional Figures**

418 Figures 9–12 show GPTZero and ZeroGPT results for both the base and instruct models, evaluated
419 on ChatGPT-sourced steered outputs (5 prompts per condition). GPTZero was evaluated manually
420 via web interface due to API cost constraints and so was not used for subsequent experiments.

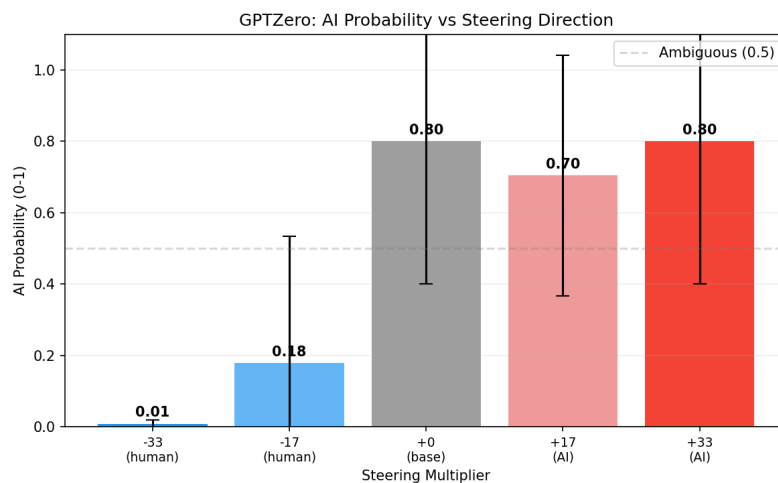


Figure 9: GPTZero AI probability by steering multiplier for Qwen2.5-3B (base). High variance across prompts and the non-monotonic positive multipliers suggest instability at 5 prompts.

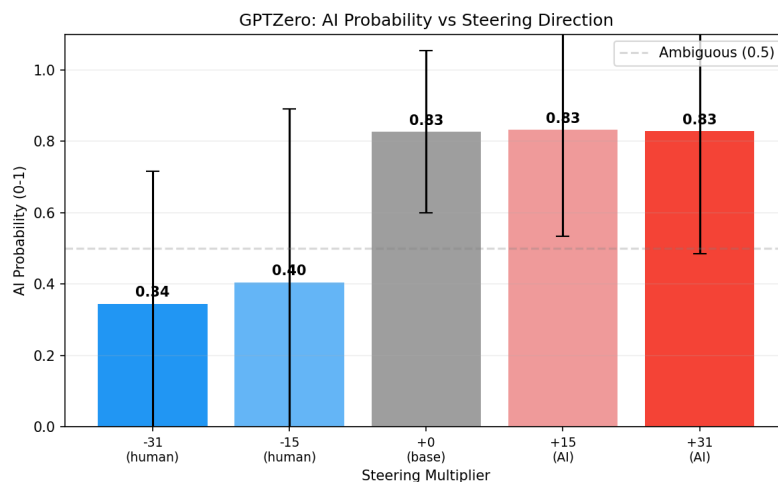


Figure 10: GPTZero AI probability by steering multiplier for Qwen2.5-3B-Instruct. The most human-steered condition reaches 0.34, but scores are flat from baseline onward (0.83 across all positive multipliers), suggesting GPTZero is largely insensitive to AI-ward steering on the instruct model.

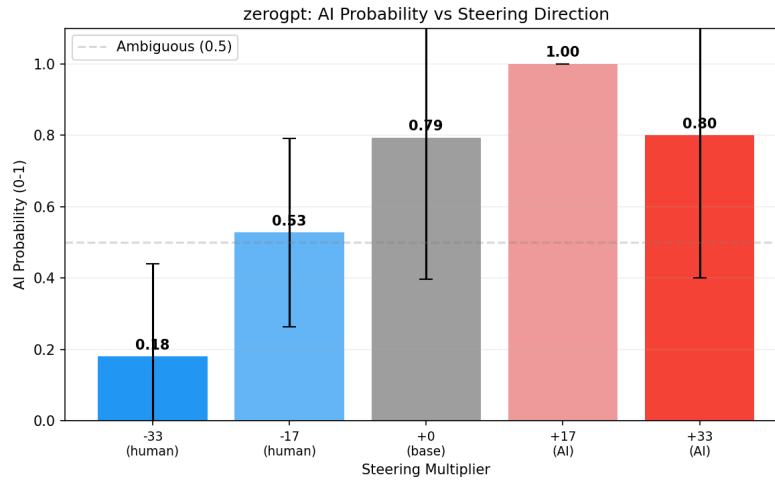


Figure 11: ZeroGPT AI probability by steering multiplier for Qwen2.5-3B (base). Results are non-monotonic: the most human-steered condition reaches 0.18, but the less human-steered condition spikes to 1.00, and the most AI-steered condition drops to 0.80. ZeroGPT appears unreliable on these short outputs (under 150 words), consistent with the detector’s own guidance on minimum input length.

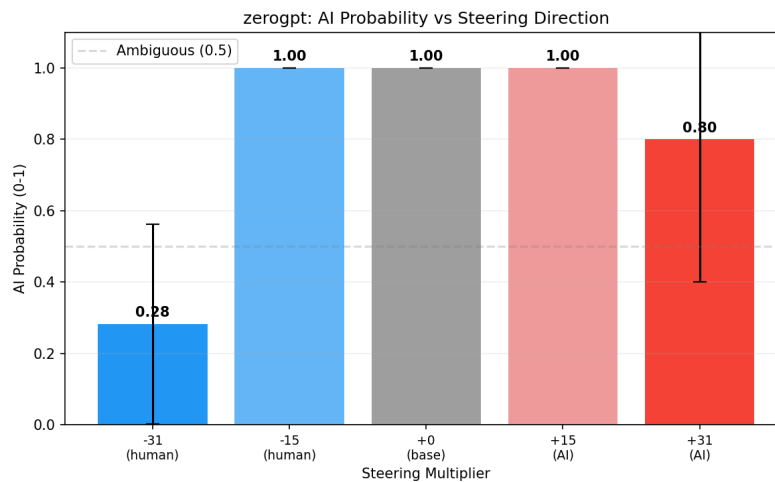


Figure 12: ZeroGPT AI probability by steering multiplier for Qwen2.5-3B-Instruct. The most human-steered condition reaches 0.28, but the pattern is again non-monotonic, with the less human condition jumping to 1.00 and the most AI condition dropping to 0.80. As with the base model, results are unreliable at this output length.

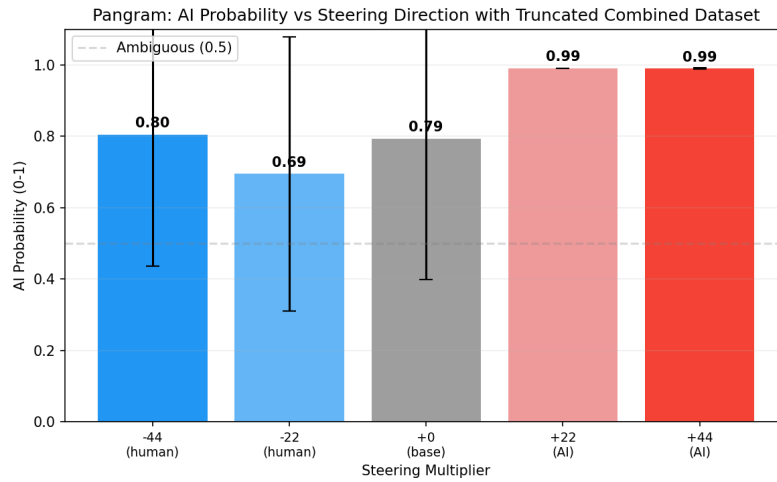


Figure 13: Pangram AI probability by steering multiplier on the truncated pooled dataset, evaluated over 5 prompts. The direction appears reversed relative to the ChatGPT-only result, with negative (human-steered) multipliers scoring higher than positive (AI-steered) multipliers. As shown in Section 5.6, this reversal disappears when evaluated over 30 prompts, indicating it is a sampling artifact rather than a true property of the pooled direction.